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CIS 344

Project 1

Sept 29th, 2026

Online Career Coaching System

Gather requirements

The requirements for the Online Career Coaching System were identified by examining how an online career coaching service would work in a real-world situation. The main users of the system are clients and career coaches. The system needs to allow clients to create accounts, search for coaches, view their areas of expertise, schedule coaching sessions, make payments, and leave reviews. Career coaches need to be able to create profiles, provide information about their professional expertise, manage coaching sessions, and receive reviews from clients. Based on these requirements, the system will store information about clients, coaches, areas of expertise, coaching sessions, payments, and reviews. These requirements will be used to determine the entities, attributes, and relationships needed for the ER diagrams and database.

Client requirements:

- A client can create an account.
- A client can maintain their personal information.
- A client can search for career coaches.
- A client can view a coach's expertise.
- A client can schedule coaching sessions.
- A client can make payments for sessions.
- A client can leave a review after a coaching session.

Career Coach requirements:

- A coach can create a profile.
- A coach can provide their professional information.
- A coach can have one or more areas of expertise.
- A coach can offer coaching sessions.
- A coach can manage scheduled sessions.
- A coach can receive reviews from clients.

System requirements:

- The system must store client information.
- The system must store coach information.
- The system must store areas of expertise.
- The system must store scheduled sessions.
- The system must store payment information.
- The system must store client reviews.

Documented Process / How Requirements Were Determined

The requirements for the Online Career Coaching System were determined by examining how an online career coaching service would work in a real-world situation. The main users of the system are clients and career coaches. I considered the activities that each user would need to perform and the information the system would need to store.

First, I identified the main users of the system. Clients are people who use the service to find career guidance, while career coaches provide professional advice and support. From these users, I identified the main functions that the system should provide, such as creating accounts, managing profiles, searching for coaches, and scheduling coaching sessions.

Next, I identified the information that needs to be stored for each user. Clients need information such as their name, email, and contact information. Career coaches need information about their name, contact information, professional background, and areas of expertise. This information would allow clients to find coaches based on their career needs.

I then identified the activities that occur between clients and career coaches. A client should be able to select a coach and schedule a coaching session based on the coach's availability. The system should store the date, time, and status of each session. This created the need for a Coaching Session entity that connects clients and career coaches.

After identifying the scheduling process, I determined that the system should also keep track of payments for coaching sessions. Payment information would include the amount, payment date, payment method, and payment status. This allows the system to keep a record of transactions associated with coaching sessions.

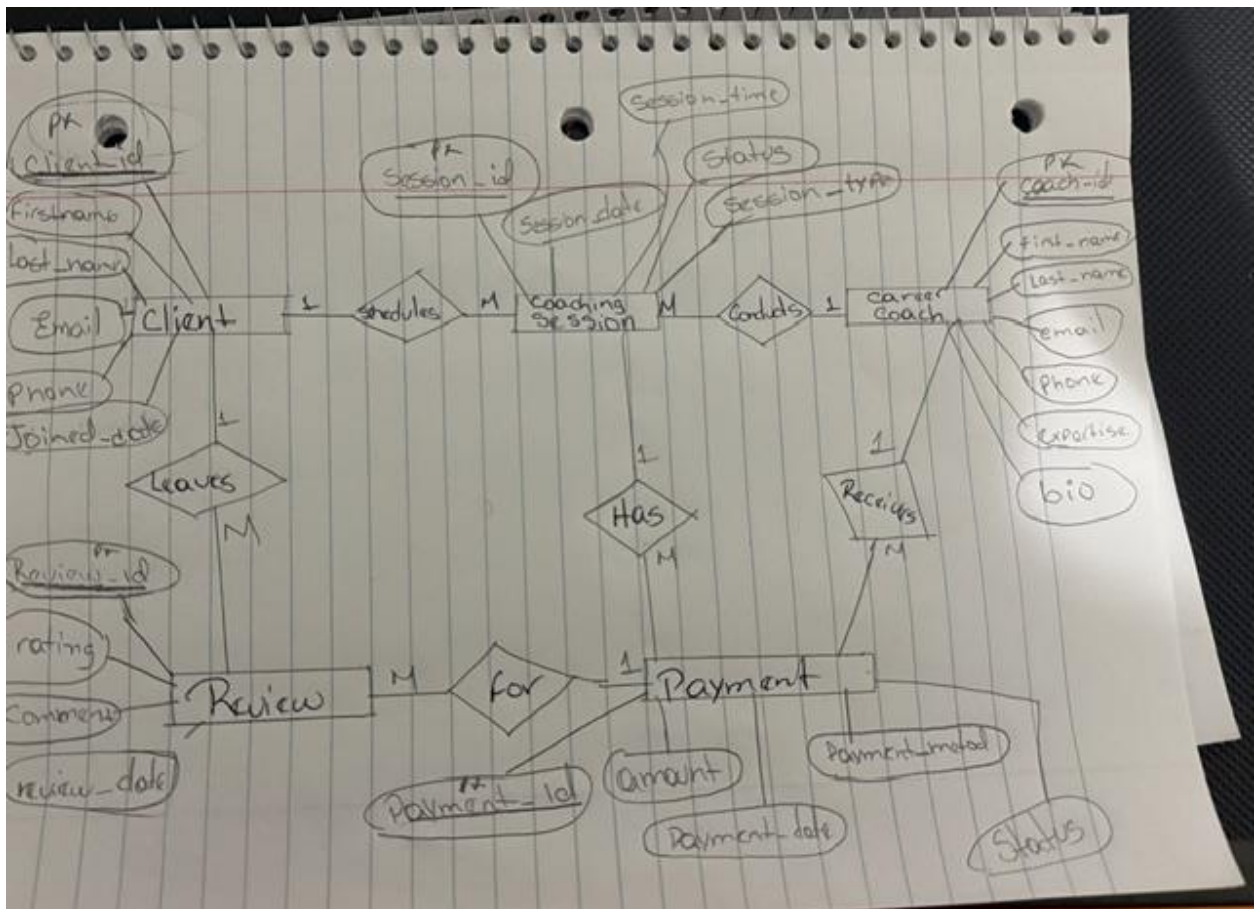
I also considered how clients could provide feedback after receiving coaching services. The system should allow clients to submit reviews and ratings for career coaches. Reviews would include information such as the rating, comments, and date of the review. This information can help other clients learn about a coach's services.

Finally, I reviewed the requirements to make sure that the entities and relationships represented the main activities of the system. The requirements were organized into

clients, career coaches, coaching sessions, payments, and reviews. These requirements will be used to create the Chen-style ER diagram, the UML ER diagram in MySQL Workbench, and the database tables. This process helped ensure that the database design represents the main functions and information needed for an Online Career Coaching System.

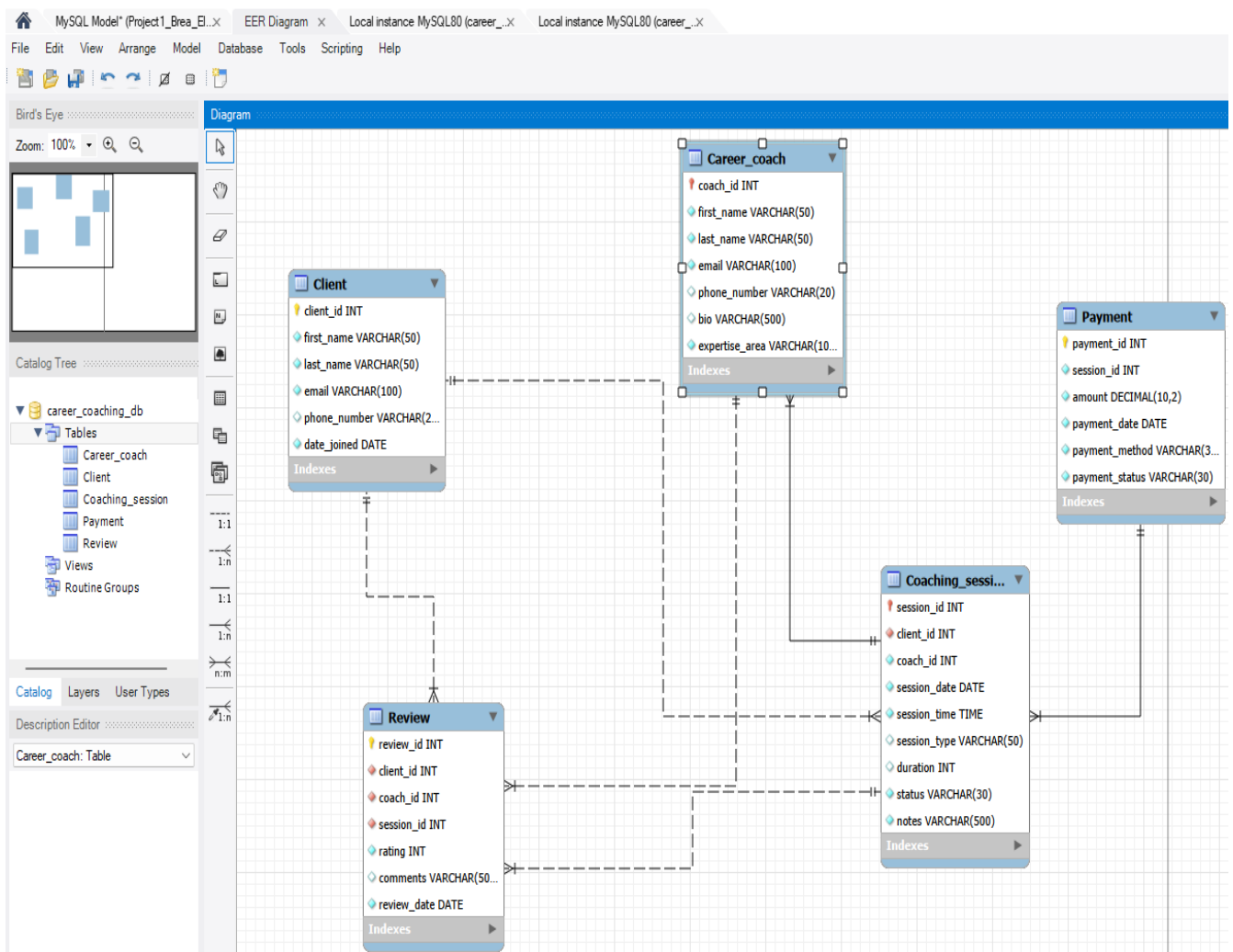
Chen Style ER Diagram

The Chen Style ER Diagram was drawn by hand based on the requirements gathered for the Online Career Coaching System. The diagram represents the main entities, their attributes, primary keys, relationships, and cardinalities within the system.



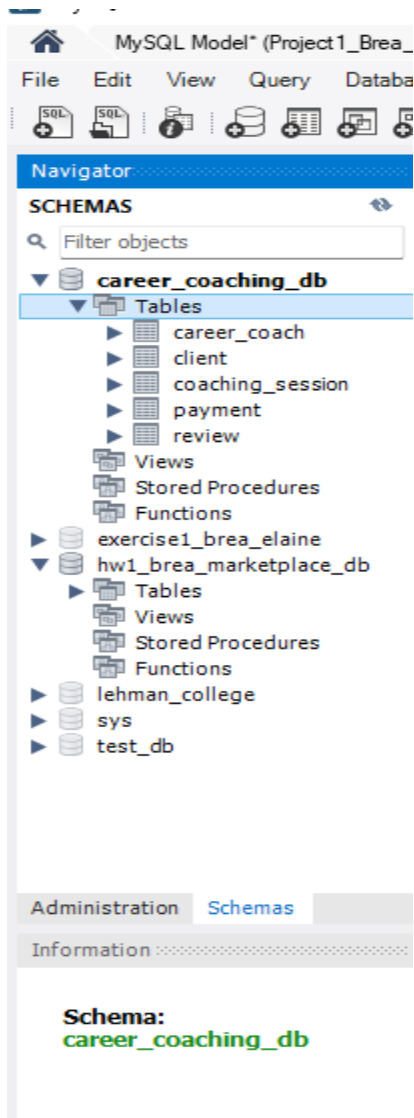
UML Style ER Diagram

The UML Style ER Diagram was created using MySQL Workbench based on the requirements identified for the Online Career Coaching System. The diagram contains five main tables: Client, Career Coach, Coaching Session, Payment, and Review. Primary keys are used to uniquely identify each record, while foreign keys are used to create relationships between the tables. The Coaching Session table connects clients with career coaches, while the Payment table records payments associated with coaching sessions. The Review table connects clients, career coaches, and coaching sessions. The UML diagram was also used as the foundation for creating the actual database in MySQL Workbench.



Database Implementation

The Online Career Coaching System database was created using MySQL Workbench. The database was named `career_coaching_db` and contains five tables: Client, Career Coach, Coaching Session, Payment, and Review. Each table has a primary key that uniquely identifies each record. Foreign keys were used to connect related tables and maintain relationships between the data. The Coaching Session table connects clients with career coaches. The Payment table stores payment information for coaching sessions, while the Review table stores feedback provided by clients about their coaching experience. The database was created using the Forward Engineer feature in MySQL Workbench. Sample records were also added to each table to test the database and confirm that the relationships work correctly.



SQL Implementation

SQL was used to create and manage the Online Career Coaching System database. MySQL Workbench generated the SQL statements needed to create the database, tables, primary keys, and foreign key relationships. SQL INSERT statements were also used to add sample data for clients, career coaches, coaching sessions, payments, and reviews. SELECT statements were used to view the stored data and verify that the records were inserted correctly. The completed database structure and sample data were exported into the Project1_Brea_Elaine.sql file. This SQL file can be used to recreate the database and its data.

The screenshot displays the MySQL Workbench interface. The top toolbar includes icons for file operations, execution, and navigation. Below the toolbar, a list of SQL queries is shown, each preceded by a blue bullet point:

- 2
- 3 • `SELECT * FROM Client;`
- 4 • `SELECT * FROM Career_coach;`
- 5 • `SELECT * FROM Coaching_session;`
- 6 • `SELECT * FROM Payment;`
- 7 • `SELECT * FROM Review;`

The 'Result Grid' tab is active, showing the results of the last query. The grid has columns: review_id, client_id, coach_id, session_id, rating, comments, and review_date. The data is as follows:

review_id	client_id	coach_id	session_id	rating	comments	review_date
1	1	4	13	5	Great interview preparation session.	2026-09-15
2	2	5	14	5	Very helpful resume advice.	2026-09-18
3	3	6	15	4	Helpful career planning session.	2026-09-25
* NULL	NULL	NULL	NULL	NULL	NULL	NULL

Below the result grid, the 'Output' tab is active, showing the 'Action Output' for the executed queries. The output is as follows:

#	Time	Action	Message
71	17:26:47	USE career_coaching_db	0 row(s) affected
72	17:26:47	SELECT * FROM Client LIMIT 0, 1000	3 row(s) returned
73	17:26:47	SELECT * FROM Career_coach LIMIT 0, 1000	3 row(s) returned
74	17:26:48	SELECT * FROM Coaching_session LIMIT 0, 1000	3 row(s) returned
75	17:26:48	SELECT * FROM Payment LIMIT 0, 1000	3 row(s) returned
76	17:26:48	SELECT * FROM Review LIMIT 0, 1000	3 row(s) returned

Challenges and Solutions

During the development of the Online Career Coaching System, I experienced some challenges while creating the relationships between the tables. Some foreign keys were initially connected to the wrong tables, which caused errors when sample data was inserted. I reviewed the foreign key settings in MySQL Workbench and corrected the relationships so that each foreign key referenced the correct primary key. I also encountered duplicate records while testing the SQL statements. These duplicate records were removed using DELETE statements, and the data was checked again using SELECT statements. After correcting these issues, the database successfully stored clients, career coaches, coaching sessions, payments, and reviews. This process helped me better understand how primary keys, foreign keys, and table relationships work in a relational database.

Conclusion

The Online Career Coaching System was successfully designed and implemented using MySQL Workbench. The final database contains five main tables: Client, Career Coach, Coaching Session, Payment, and Review. These tables work together to store information about clients, coaches, scheduled sessions, payments, and client feedback. Primary keys and foreign keys were used to organize the data and establish relationships between the tables. Sample data was inserted to test the database and verify that the relationships worked correctly. The project provided experience with gathering system requirements, creating Chen and UML ER diagrams, designing relational database tables, and using SQL to create and manage data. Overall, the completed database meets the main requirements identified for the Online Career Coaching System.